

Jon Kozlik

848-205-3464 | jmk98@njit.edu | linkedin.com/in/jkozlik | github.com/jmkozlik

EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. in Computer Science and Applied Mathematics, GPA: 3.50

Expected Dec 2027

Speaker: MIT Decentralized AI Conference 2025 (MCP security: encrypted inference and access control)

Awards: NSF S-STEM Scholar; Theodore E. Kupfrian Memorial Scholarship; HackMIT 2024 Track Runner-Up

EXPERIENCE

MIT Media Lab

Nov 2024 – May 2025

Core Developer, NANDA Protocol (Prof. Ramesh Raskar's Group)

Cambridge, MA

- Built a DNS-style registry for independently operated Model Context Protocol (MCP) servers in NANDA
- Reduced invalid and spam registrations ~70% via coupon-based ownership verification
- Improved discovery of high-quality MCP servers 40% via a community-ranking system on the registry

Southern Company

May 2026 – Present

AI Application Engineer Intern, AI Center of Excellence

Atlanta, GA

- Built an end-to-end LLM RAG platform over 100K+ sensitive documents on GCP Firestore and Google ADK
- Delivered sub-second retrieval at 100K+ document scale via metadata-aware chunking and pruning
- Implemented document-level access controls; owned the agent harness and deployment end-to-end

Encryptech AI

Oct 2024 – Nov 2025

Co-Founder & AI/ML Engineer

Newark, NJ

- Won a \$50K non-dilutive grant from Codebase by Avalanche as 1 of 5 teams from ~100 applicants
- Trained models with DP-SGD under enforced privacy budgets, limiting validation-accuracy loss to under 4%
- Implemented homomorphically encrypted federated aggregation across 3+ silos, no raw records exchanged

Trainsafe Inc.

2025 – Present

Co-Founder and Lead Engineer

Newark, NJ

- Built a multi-tenant agent-workflow SaaS generating PowerPoint, Word, and Excel deliverables via LLMs
- Designed an async workflow engine: 19 typed nodes, resumable human-in-the-loop pauses, and a dual-store memory separating model-visible context from node outputs
- Served published workflows as MCP tools via a JSON-RPC endpoint secured with HMAC-scoped tokens

PROJECTS

Constitutional Multi-Agent Development System | Python, Git worktrees

2026

- Compiled a core constitution into 12 role charters governing executor agents in isolated Git worktrees
- Executed 12+ contracted dev tasks across 11 governed runs (164-entry audit ledger); adversarial verifiers caught 10+ defects missed by passing test fixtures
- Fed failures back into the law: verifier-caught defects filed as cited motions, ratified into amended contracts

macrograd | Python (no NumPy or PyTorch), Graphviz

2026

- Built a matrix autograd engine from scratch: hand-derived matmul, broadcasting, and loss gradients
- Implemented topological-sort backprop, validating gradients against finite differences to ~3e-10 max error

SmartFarm (FarmIQ) | C++, ESP32, MQTT, MapLibre | Cox Enterprises Hackathon

2025

- Built a C++ ESP32 sensor mesh over ESP-NOW (220m per node) driving automated MQTT pump control
- Shipped a live satellite overlay of the field that aggregates decentralized per-patch soil readings
- Modeled ~38% potential water savings (~16M gal/season, 100-acre farm) via FAO-56 dry-down modeling

SKILLS

Languages: Python, C, C++, Bash, JavaScript (Node.js), SQL

AI & ML: Multi-Agent Systems, Agent Evaluation, LLM Orchestration, MCP, RAG, PyTorch

Privacy & Security: DP-SGD, Federated Learning, Homomorphic Encryption, Access Control

Tools & Infrastructure: GCP, Firestore, Google ADK, FastAPI, asyncio, ESP32, MQTT, Git, Docker, Railway